

Molecular Photocopying

How Polymerase Chain Reaction (PCR) Amplifies the Code of Life

THE BIG IDEA

PCR selectively amplifies a specific DNA sequence, turning a small genetic signal into enough material for analysis.

Think of it as a molecular photocopier: identify one sentence in an enormous library, then repeatedly copy it until it becomes easy to study.

DNA
TARGET

1

AMPLIFY

Specific target DNA

2

ANALYSE

Gel, qPCR or sequencing

3

APPLY

Diagnostics • AMR • forensics

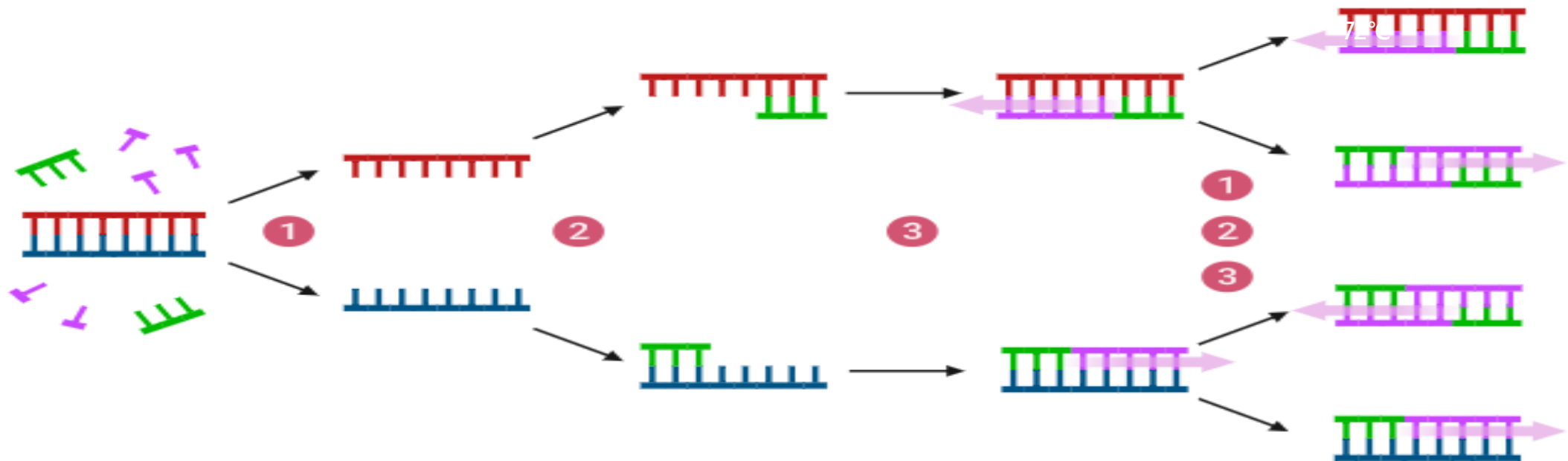
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The PCR cycle: three controlled temperature steps

A thermal cycler repeatedly changes temperature so DNA can separate, primers can bind and polymerase can synthesise new strands.

Polymerase chain reaction - PCR

Each cycle provides new templates for the next round.



1) Denaturing (95-96°C)

Heat disrupts the hydrogen bonds between complementary bases, separating double-stranded DNA into single templates.

2) Annealing (68°C)

Cooling allows short DNA primers to bind to complementary sequences flanking the target region. Primer sequence and temperature help determine specificity.

3) Elongation (72°C)

A thermostable DNA polymerase extends from the primers using dNTPs. Conventional PCR commonly uses Taq polymerase.

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Exponential amplification—and why specificity matters

In an ideal reaction, target DNA approximately doubles each cycle: 2^n , where n is the number of cycles.



WHAT MAKES PCR SPECIFIC?

Primer pair	defines the target region and provides the starting point for synthesis.
Reaction conditions	including annealing temperature, help minimise non-specific amplification.
Controls	positive and negative controls help distinguish genuine amplification from contamination.

AFTER AMPLIFICATION

PCR products can be visualised by agarose gel electrophoresis, quantified in qPCR workflows or prepared for sequencing.

Real reactions are not perfectly efficient and eventually reach a plateau as reagents become limiting.

Why PCR matters in biomedical science

01

INFECTIOUS-DISEASE DIAGNOSTICS

Targeted amplification can identify pathogen nucleic acid sequences. RNA targets, such as SARS-CoV-2, require an additional reverse-transcription step before amplification.

02

ANTIMICROBIAL-RESISTANCE RESEARCH

PCR can screen clinical or environmental samples for genes associated with antimicrobial resistance, including targets investigated in water and wastewater surveillance.

03

FORENSIC DNA PROFILING

Amplification enables analysis of selected DNA regions from very small biological samples, supporting genetic profiling.

04

GENETIC & GENOMIC RESEARCH

PCR supports cloning, sequencing preparation and targeted genetic analysis, often forming one stage of a larger experimental workflow.

From technique to reliable biomedical evidence

WHAT THIS TECHNIQUE DEMONSTRATES

- **Sequence specificity** Primers define what is amplified.
- **Experimental control** Positive and negative controls support interpretation.
- **Reproducibility** Temperature, reagent choice and cycle number affect performance.
- **Downstream analysis** Amplified products must be visualised, quantified or sequenced.

KEY TAKEAWAY

PCR is not simply a way to make “more DNA” . It a sequential amplification system linking molecular biology, experimental design and downstream analysis.

Understanding primer specificity, thermal cycling, polymerase activity and controls is fundamental to using PCR reliably in a biomedical laboratory.

KEY REFERENCES

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Saiki, R.K. et al. (1988). Primer-directed enzymatic amplification of DNA with a thermostable DNA polymerase. Science, 239(4839), 487–491.

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